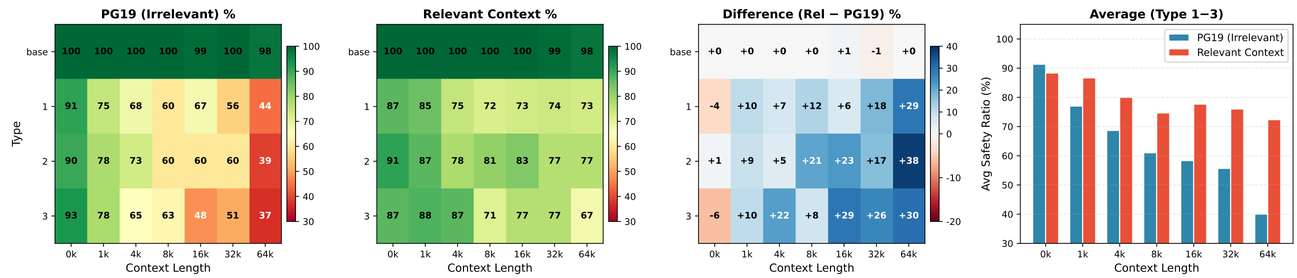

Figure 6. Safety ratio comparison between irrelevant context (PG19 books) and relevant context (topic-related background). The difference heatmap shows the improvement from relevant context, and the bar chart (right) summarizes the average safety rates across Type 1–3.

The Dilemma of Static Reasoning Effort. Although the efficiency trade-off suggests a theoretical "sweet spot" for reasoning effort, identifying this optimum in practice is challenging. As illustrated in Figure 4, the marginal utility of additional reasoning tokens varies drastically across effort levels, with high efficiency at lower effort but diminishing returns at higher effort. This non-linearity creates a fundamental dilemma: a static reasoning-effort setting either incurs prohibitive computational cost through over-reasoning or risks severe safety failures due to under-reasoning.

3.3.2. Task Complexity Analysis

Next, we examine how task complexity influences safety alignment (Figure 5). Under low reasoning effort, safety decreases as task complexity increases: Single-Hop achieves 31%, Chain drops to 29%, and Multi-Hop further decreases to 21%. This pattern suggests a budget-allocation effect: more complex tasks may consume most of the available reasoning effort to reconstruct the harmful intent, leaving less budget for the model to recognize the harm and refuse.

Task complexity also modulates the benefits of increasing reasoning effort. As shown in Figure 5, Multi-Hop achieves a safety improvement of 277% on GPT-oss-120b when moving from low to high reasoning effort, compared to 158% for Single-Hop. Together, these results indicate that complex reasoning tasks are most vulnerable under low reasoning budgets, yet also gain the most from additional reasoning effort. This complements StrataSword (Zhao et al., 2025b), which shows that attack complexity affects safety; here we show a parallel effect on the defense side, where the impact of reasoning effort is similarly task-dependent.

3.4. Long-Context Analysis

Although we analyzed how reasoning effort and task complexity affect safety, our main results also show that context length is a critical factor. To understand why safety degrades in long contexts, we focus on two mechanisms: *context relevance* (whether safety depends on the full context or on task-relevant fragments) and *needle position* (how the location of fragments within the context affects safety).

3.4.1. Context Relevant Analysis

We study how the surrounding context influences safety by comparing two conditions: *irrelevant context* (as in Table 1, where the decomposed fragments are embedded in unrelated documents) and *relevant context* (where an uncensored LLM generates task-related neutral background text that semantically connects to the harmful query). The generation prompts for relevant context are provided in Appendix F.

As shown in Figure 6, the models achieve *higher* safety ratios when the fragments are embedded in a relevant context than in an irrelevant context. At 4K context length, the relevant context yields an average safety ratio of 80% versus 68.7% for the irrelevant context (+11.3 points). This gap grows with context length, increasing from +11.3 points at 4K to +32.3 points at 64k. This result is counter to previous work (Lu et al., 2025), which finds that task-relevant information can facilitate attack success. In our setting, the relevant context appears to provide contextual clarity: it makes the reconstructed harmful intent more explicit and coherent, which may make it easier for safety mechanisms to detect and reject. By contrast, irrelevant context can create a *camouflage effect*, where harmful intent remains diffuse and implicit amid noisy distractors, potentially slipping past safety filters.

3.4.2. Needle Position Analysis

Another question in long-context modeling is whether the position of critical information affects performance. Prior work shows a "lost-in-the-middle" effect (Liu et al., 2024), where models retrieve information from the middle of long contexts less reliably than from the beginning or end. We test whether this retrieval vulnerability translates into safety failures by systematically varying the placement of decomposed fragments across four regions: Start, Early Middle, Late Middle, and End. As shown in Figure 7, the safety ratios are insensitive to the position of the fragments in all context lengths. For GPT-oss-120b at 16k context, safety varies only within a narrow range across the four regions.

This position-invariant behavior suggests that models can

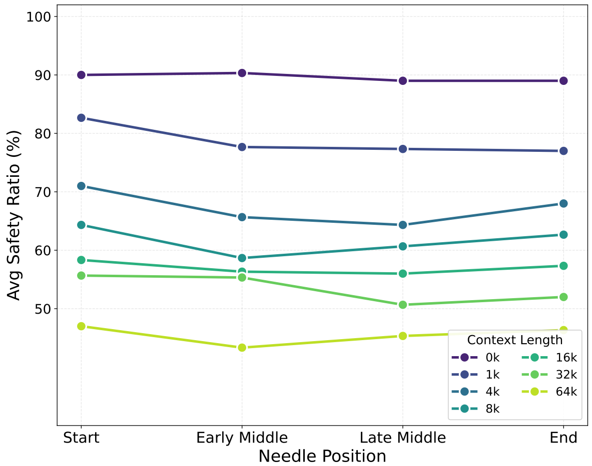

Figure 7. Safety ratio across needle positions for GPT-oss-120b. Safety remains largely stable regardless of fragment placement, indicating no "lost-in-the-middle" effect for safety.

retrieve and integrate decomposed fragments regardless of where they appear in the context. However, our main results (Table 1) still show a substantial degradation in safety as the length of the context increases. Together, these findings indicate that long-context safety failures are not primarily driven by an inability to locate relevant fragments. Instead, failures arise after successful retrieval: models reconstruct the harmful intent but do not reliably recognize it as harmful and refuse. This suggests that improving long-context safety will require strengthening intrinsic safety alignment mechanisms, rather than focusing only on retrieval or general reasoning capability.

4. Related Works

Long-Context Reasoning Benchmarks. Evaluation of long-context LLMs has progressed from retrieval to multi-step integration. Early work emphasized retrieval tasks such as needle-in-a-haystack (Li et al., 2025b), with extensions like RULER (Hsieh et al., 2024) (multi-needle variants) and NoLiMa (Modarressi et al., 2025) (reduced lexical overlap to limit pattern matching). Recent benchmarks, including LongReason (Ling et al., 2025), FanOutQA (Zhu et al., 2024), LongBench v2 (Bai et al., 2025), and BABILong (Kuratov et al., 2024), test reasoning over extended contexts and show performance degradation as context grows (Ling et al., 2025; Wang et al., 2024). However, these benchmarks measure capability: whether a model can synthesize distributed information, rather if it remains safe when synthesis reconstructs harmful intent hidden across long contexts.

Safety in Long-Context Settings. Safety work in long contexts has focused on three related issues. Many-shot jailbreaking (Anil et al., 2024) shows that attacks can scale with long contexts; follow-up work (Kim et al., 2025) argues that context length itself is the key vulnerability, with success even under meaningless or safe repetitions. Separately, alignment degradation has been documented in long-context settings: LongSafetyBench (Huang et al., 2025) reports that many long-context LLMs fall below 50% safe response rates. Finally, long-form generation raises faithfulness challenges, including hallucinations beyond the provided context (He et al., 2025) and failures under conflicting evidence (Zheng et al., 2025; Kurfali & Östling, 2025).

Safety Risks in Reasoning Models. A growing body of work treats reasoning as an attack surface. SafeChain (Jiang et al., 2025) finds that long Chain-of-Thought (CoT) traces can steer models toward compliance with harmful requests. Related attacks, including CoT Hijacking (Zhao et al., 2025a) and H-CoT (Kuo et al., 2025), show that padding harmful queries with benign reasoning can dilute safety signals. Automated frameworks such as AutoRAN (Liang et al., 2025) and Mousetrap (Yao et al., 2025) further generate prompts that leverage the model's own logic to rationalize harmful outputs, suggesting that reasoning may facilitate jailbreaking without targeted alignment (Sabbaghi et al., 2025; Ying et al., 2025). StrataSword (Zhao et al., 2025b) categorizes jailbreaks by reasoning complexity and finds that more complex attacks are generally less safe. Proposed defenses include ReasoningShield (Li et al., 2025a) and CoT-style safety training via SafeChain (Jiang et al., 2025), though SafeRBench (Gao et al., 2025) shows reasoning LLMs remain vulnerable across safety dimensions.

5. Conclusion

In this work, we investigate the safety risks arising from long-context reasoning through compositional reasoning attacks, in which harmful intent is distributed across extended contexts. Evaluating 14 frontier LLMs, we find that strong reasoning capability alone does not ensure safety alignment: models can retrieve and integrate dispersed fragments, yet often fail to refuse the reconstructed harmful intent. Moreover, safety degrades consistently as the length of the context increases, and our analysis suggests that this is not due to retrieval failures but to weaknesses in applying safety judgment after successful long context reasoning.

We identify inference-time reasoning effort as a key mitigating factor. Increasing reasoning effort within the same model improves safety, indicating that the capability to recognize harm is present but not reliably engaged by default. These findings expose a fundamental tension: the reasoning abilities that enable long-context understanding also create new attack surfaces, and mitigating these risks through increased reasoning effort incurs substantial computational cost. Addressing this challenge will require adaptive safety mechanisms that balance security and efficiency based on both model characteristics and context lengths.

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Impact Statement

This work aims to improve the safety and reliability of large language models by studying vulnerabilities that arise from long-context reasoning and compositional attacks. By systematically analyzing how harmful intent can be reconstructed from distributed fragments, our findings highlight failure modes that are not captured by standard safety evaluations focused on short or explicit prompts. Understanding these vulnerabilities is essential for deploying LLMs safely in real-world settings where models must reason over long documents and complex contexts.

As with prior research on security and safety, our results have a dual-use nature. Insights into compositional attack strategies and long-context failure modes could potentially be misused to bypass existing safety mechanisms. However, we argue that rigorously identifying and characterizing these weaknesses is a necessary step toward building more robust defenses. By showing that safety failures persist even when retrieval and reasoning succeed, our work points to the need for adaptive safety mechanisms that scale with reasoning depth and context length. Overall, we expect this work to have a positive societal impact by informing the design of safer long-context LLMs, guiding future safety evaluations beyond surface-level benchmarks, and encouraging further research into defense strategies that remain effective as models' reasoning capabilities continue to improve.

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